5.3 Local Extrema and Concavity

Name _____ Period ____ Date _____

AP Calculus 5.3 · Student sheet

I can…

- use the First Derivative Test to find local extrema.
- find where a graph is concave up or down and find inflection points.
- use the Second Derivative Test.

Vocabulary · fill in the missing word

First Derivative Test — if f' _____ from + to − at c, f(c) is a local max; from − to +, a local min

Concave up — $f'' > 0$: the graph bends _____ (its slopes increase)

Inflection point — a point on the graph where the _____ changes

Second Derivative Test — $f'(c) = 0$ and $f''(c) < 0$: _____ max; $f''(c) > 0$: local min

Part 1 · Smile or frown

$f(x) = x^3 - 3x$: slope of the tangent line.

x	−2	−1	0	1	2
Slope					

Part 2 · Notes and examples (fill in each note, then work the example)

First Derivative Test: if f' changes from positive to negative at c, then f(c) is a local _____.

Example 1

Find the local extrema of $f(x) = x^3 - 3x^2 - 9x + 1$.

Concavity: if $f''(x) > 0$ on an interval, the graph is concave _____ there.

Example 2

Where is $f(x) = x^3 - 3x^2 - 9x + 1$ concave up? Concave down?

Inflection point: a point where the _____ changes.

Example 3

Find the inflection point of $f(x) = x^3 - 3x^2 - 9x + 1$.

Second Derivative Test: if $f'(c) = 0$ and $f''(c) < 0$, then f(c) is a local _____.

Example 4

Use the Second Derivative Test on $f(x) = x^4 - 4x^3$.

Example 5

$f'(x) = (x - 1)^2(x + 2)$. Find the local extrema of f.

Part 3 · Partner practice

Partner 1

Where does $f(x) = x^3 - 12x$ have a local maximum?

Partner 2

(AP® Practice) The graph of $f(x) = x^3 - 6x^2 + 5$ has an inflection point at $x =$

(A) 2 (B) 0 (C) 4 (D) 6